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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

IS2213-1 – OBJECT ORIENTED MODELING AND DESIGN
(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is a key feature of Object Orientation?
A) Compilation B) Abstraction
C) Debugging D) Modular Programming
2. What does OO development primarily focus on?
A) Algorithms B) Procedural programming
C) Object and class-based design D) Machine-level programming
3. What is the purpose of OO modeling?
A) Hardware simulation B) System abstraction
C) Compiler design D) Cache optimization
4. The origin of OO can be traced to _____.
A) FORTRAN B) C
C) Simula D) Lisp
5. Reification means:
A) Ignoring associations B) Creating classes from relationships
C) Deleting abstract classes D) Renaming methods
6. Constraints in UML are shown using:
A) Curly braces {} B) Square brackets []
C) Parentheses () D) Dashes --
7. A self-transition:
A) Does not exist B) Moves to parent state
C) Goes back to the same state D) Deletes the object
8. Nested states in a state diagram are used to:
A) Add attributes B) Show object hierarchy
C) Group related substates D) Define packages
9. Signal generalization refers to:
A) Creating a signal hierarchy B) Multiplying transitions
C) Compressing signal names D) Deleting signals
10. The first step in the software development life cycle is:
A) Testing B) Maintenance
C) System conception D) Coding
11. A development life cycle includes:
A) Concept only B) Design only
C) All stages from conception to maintenance D) Just modeling
12. A final state in UML state diagrams is represented by:
A) Filled circle B) Hollow circle
C) Circle with a border D) Bullseye (black circle within a white circle)

13. Which of these supports modeling object lifecycles?
 A) Sequence diagram B) Activity diagram
 C) State diagram D) Deployment diagram
14. Which is used to represent iteration in a sequence diagram?
 A) Dashed arrow B) A loop frame
 C) Swimlane D) Fork node
15. A reuse plan in system design helps in:
 A) Copying code B) Improving maintainability
 C) Avoiding inheritance D) Debugging faster
16. Which of the following is not a concern of system design?
 A) Identifying subsystems B) Deciding user font colors
 C) Allocating global resources D) Choosing control strategies
17. Which diagram helps in visualizing subsystems?
 A) Activity diagram B) Deployment diagram
 C) Component diagram D) Sequence diagram
18. Architecture of ATM system shows:
 A) GUI buttons B) Real-world use of design strategies
 C) AI planning D) Testing flow
19. Which design concern involves handling cases like "insufficient balance" or "invalid PIN"?
 A) Resource allocation B) Trade-off management
 C) Handling boundary conditions D) Reification
20. The concept of modularity in system design supports:
 A) Better encryption B) Easier maintenance and reusability
 C) Higher memory usage D) Slower execution

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is Object Orientation? Explain the key concepts of OO development.	8	L*2	1	1
b)	Explain the following terms with suitable examples Links, Associations, Multiplicity and Association End Names.	8	L2	2	1
2. a)	Explain the concept of inheritance in the context of graphic figures. How do subclasses inherit attributes, operations, and associations from their super classes? Illustrate your answer with a suitable diagram.	8	L2	1	1
b)	Differentiate between Association and Aggregation in object-oriented programming. Also, explain how Aggregation differs from Composition with appropriate examples.	8	L2	2	1
3. a)	Draw a class model for a simple windowing system. Your model should include classes such as Window, Button, TextWindow, and Panel. Indicate the relationships (like association, generalization, multiplicity etc) between the classes. Briefly explain the key features of your model.	8	L2	1	1
b)	Explain the different types of Constraints on Generalization Sets with examples.	8	L2	2	1
Unit – II					
4. a)	Draw and explain a state diagram representing a phone line system using nested states.	8	L2	3	1
b)	Explain the key guidelines to be followed while creating a Use Case Model in UML.	8	L2	4	1

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| 5. a) | What is activity diagram? Draw and explain an activity diagram for a stock trade processing system. | 8 | L2 | 3 | 1 |
| b) | Describe the main stages of the software/system development process. For each stage, explain its purpose and key activities involved. | 8 | L2 | 4 | 1 |
| 6. a) | Define a Sequence Diagram. Illustrate a sequence diagram for an online stock trading session between a user and a stock broker system, showing the interaction flow for placing a trade. | 8 | L2 | 3 | 1 |
| b) | Explain the process of finding new system concepts during the initial phase of system development. Discuss the techniques used to identify user requirements and system functionalities. | 8 | L2 | 4 | 1 |
| Unit – III | | | | | |
| 7. a) | Discuss the key factors to consider when choosing a software control strategy for a project. | 8 | L2 | 5 | 1 |
| b) | With a neat diagram explain the ATM domain class model. | 8 | L2 | 5 | 1 |
| 8. a) | List and Explain all the steps involved in a Class design. | 8 | L2 | 5 | 1 |
| b) | Discuss the Common Architectural Styles. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
